

Statistically Valid Inferences from Privacy Protected Data¹

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Solving Political Problems Technologically

Differential Privacy & Inferential Validity

A General Purpose, Statistically Valid DP Algorithm

The Algorithm in Practice

Convincing Facebook to Make Data Available

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Solving a Political Problem Technologically (via “constitutional design”)

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- [New Problem](#): **Sharing data without it leaving Facebook**

Data Sharing Regime $\leadsto$ Data Access Regime

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 - *no* uncertainty estimates

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Theories of Inference: Statistics vs. CS

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Population

⋮

John

Heather

Georgie

Gary

Meg

Abhradeep

Ella

Anya

Greg

Juan

Mean
income:

\$48

Quantity
of Interest

Theories of Inference: Statistics vs. CS

Population	Sample
:	X
John	✓
Heather	✓
Georgie	✓
Gary	✓
Meg	✓
Abhradeep	✓
Ella	✓
Anya	✓
Greg	✓
Juan	✓

Mean
income:

\$48

Quantity
of Interest

Theories of Inference: Statistics vs. CS

Population	Sample	\$
:	X	
John	✓	76
Heather	✓	96
Georgie	✓	145
Gary	✓	122
Meg	✓	86
Abhradeep	✓	127
Ella	✓	72
Anya	✓	132
Greg	✓	95
Juan	✓	134

Mean
income:

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Classical
Inference

\$108

Quantity
of Interest

Usually
no direct
relevance

Theories of Inference: Statistics vs. CS

Population	Sample	\$
:	X	
John	✓	76
Heather	✓	96
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Gary	✓	122
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Mean
income:

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Classical
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Quantity
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Theories of Inference: Statistics vs. CS

	Population	Sample	\$	+Privacy
	:	X		
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	Heather	✓	96	
	Georgie	✓	145	
	Gary	✓	122	
	Meg	✓	86	
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Mean income:	\$48		\$108	Noise & Censoring

Classical Inference

Quantity of Interest

Usually no direct relevance

Theories of Inference: Statistics vs. CS

	Population	Sample	\$	+Privacy	=dp\$
	:	X			
	John	✓	76	Noise & Censoring	85
	Heather	✓	96		103
	Georgie	✓	145		75
	Gary	✓	122		113
	Meg	✓	86		125
	Abhradeep	✓	127		97
	Ella	✓	72		101
	Anya	✓	132		128
	Greg	✓	95		83
	Juan	✓	134		201
Mean income:	\$48		\$108		\$111
	Quantity of Interest	Classical Inference	Usually no direct relevance	Query-Response	No direct relevance

Theories of Inference: Statistics vs. CS

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Juan	✓	134		201

Mean
income:



Differential Privacy and its Inferential Challenges

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- Estimators

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 - Classical Statistics: Apply statistic s to dataset D , $s(D)$

Differential Privacy and its Inferential Challenges

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- Statistical properties: usually biased, no uncertainty estimates

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 - **Can address with:** careful software design & education

Solving Political Problems Technologically

Differential Privacy & Inferential Validity

A General Purpose, Statistically Valid DP Algorithm

The Algorithm in Practice

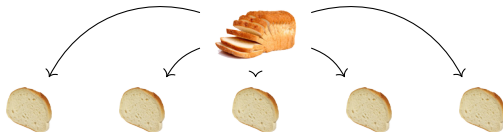
A Differentially Private Estimator

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Private data

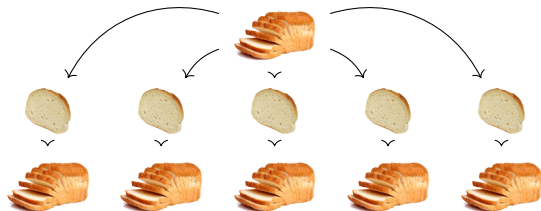
A Differentially Private Estimator



Private data

Partition

A Differentially Private Estimator

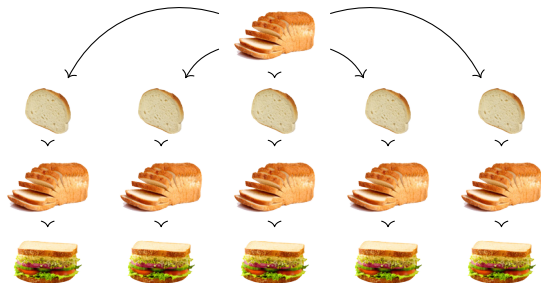


Private data

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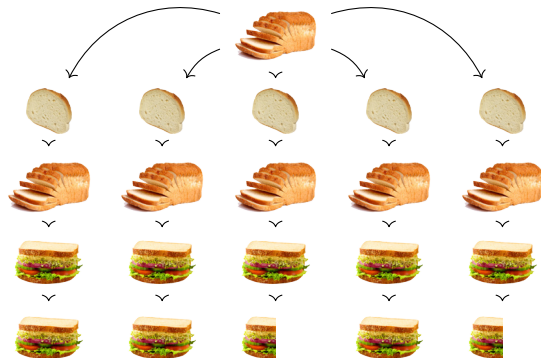
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Estimator

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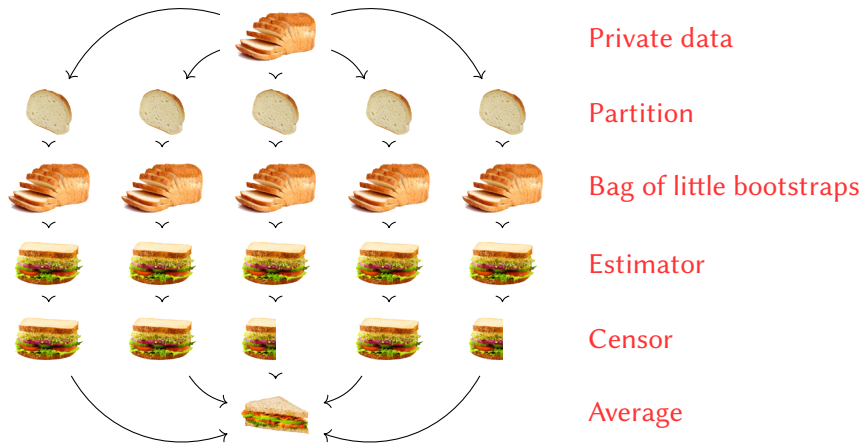
Partition

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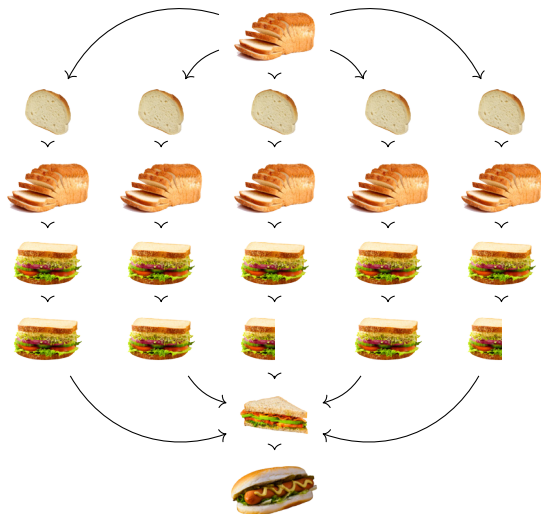
Estimator

Censor

A Differentially Private Estimator



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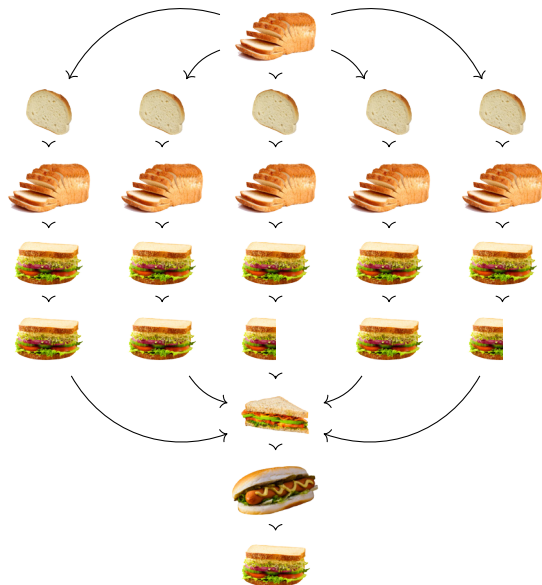
Estimator

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Average

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A Differentially Private Estimator



Private data

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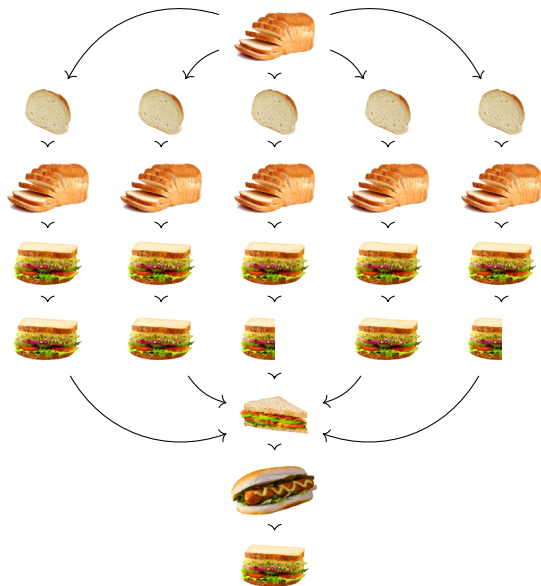
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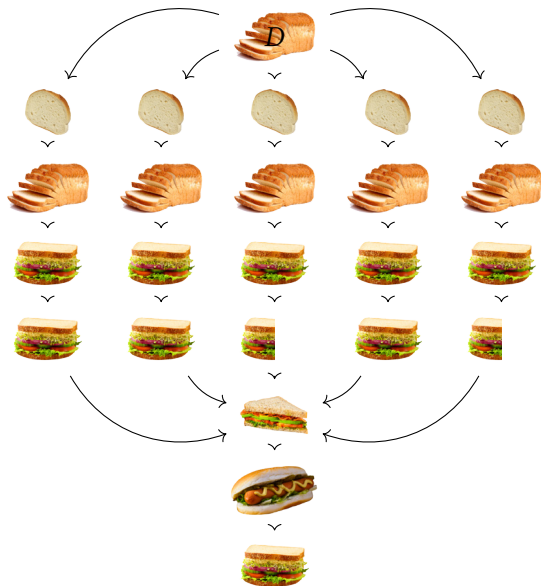
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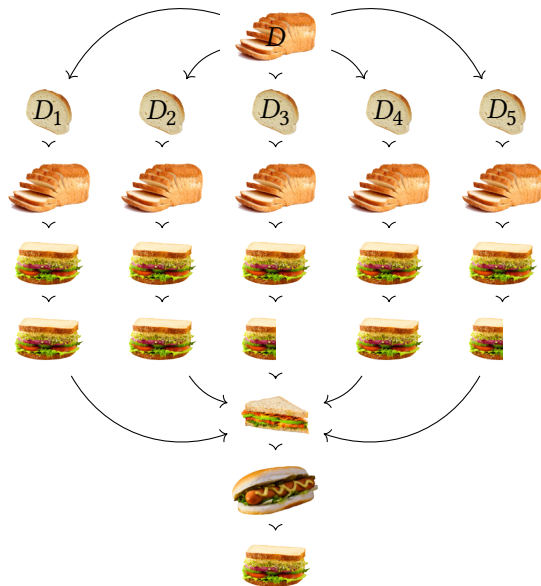
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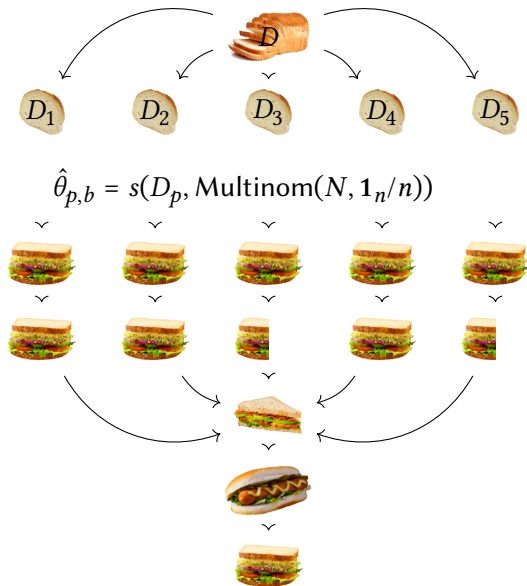
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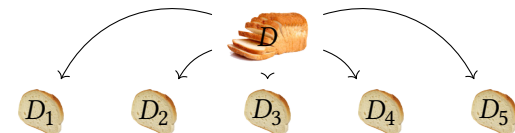
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$$\hat{\theta}_{p,b} = s(D_p, \text{Multinom}(N, \mathbf{1}_n/n))$$

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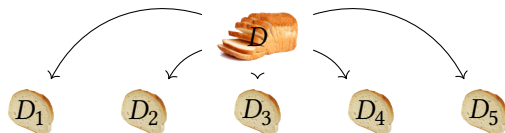


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$$\hat{\theta}_1 \quad \hat{\theta}_2 \quad \hat{\theta}_3 \quad \hat{\theta}_4 \quad \hat{\theta}_5$$

Estimator

$$\hat{\theta}^{\text{dp}} = \frac{1}{P} \sum_{p=1}^P c(\hat{\theta}_p, \Delta) + N\left(0, \frac{4\Delta}{\epsilon P}\right)$$

Censor

Average

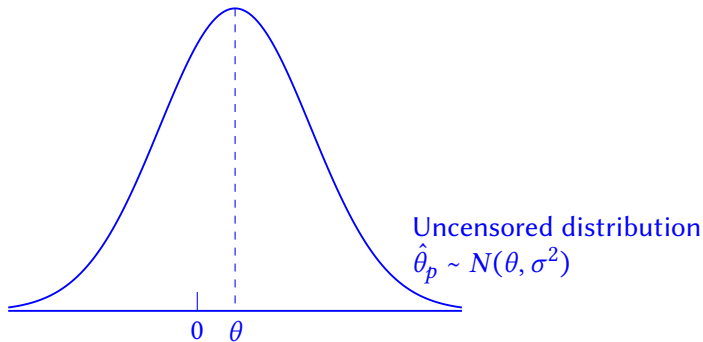
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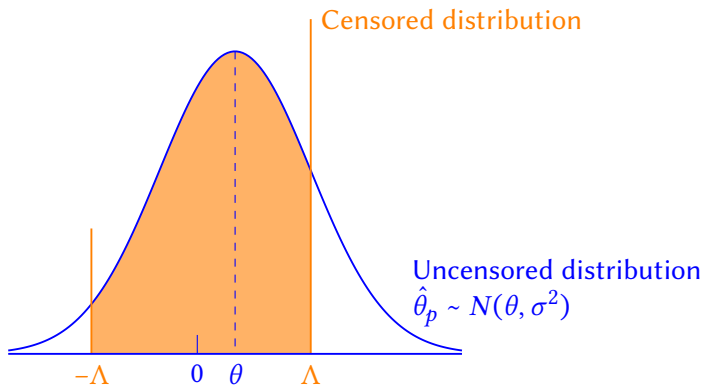
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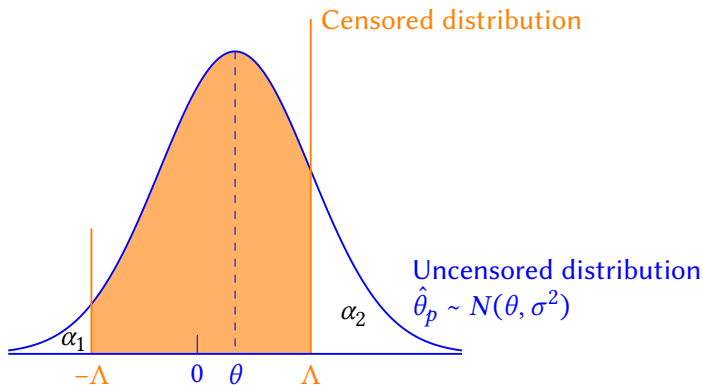
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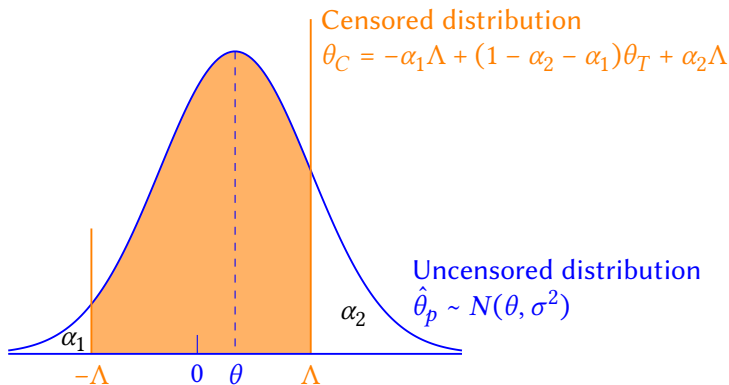
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$$\int_{-\infty}^{-\Lambda} N(t \mid \theta, \sigma^2) dt$$

$$\int_{\Lambda}^{\infty} N(t \mid \theta, \sigma^2) dt$$

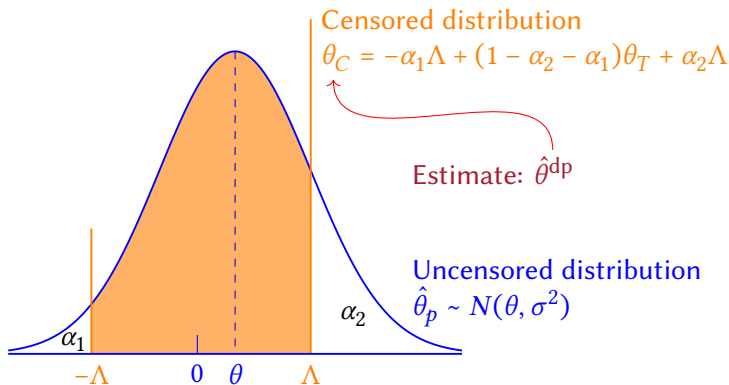
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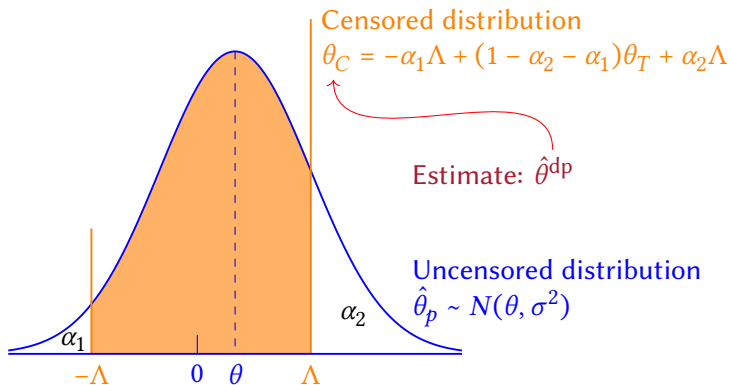
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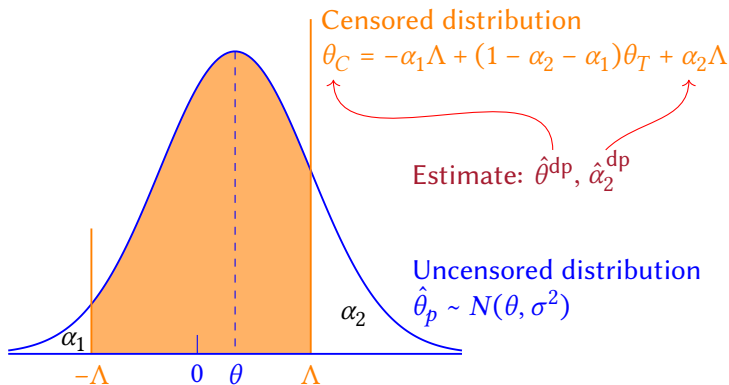


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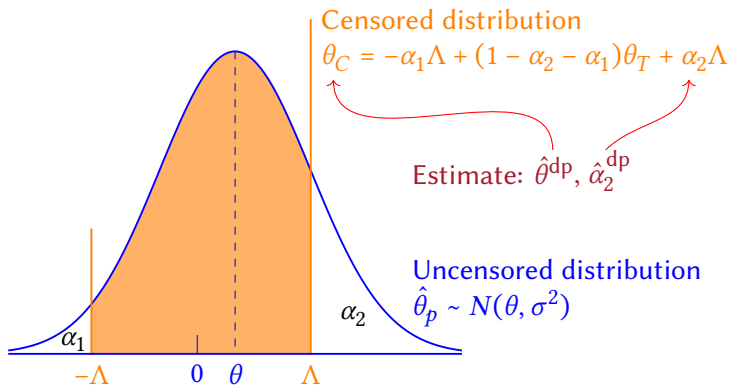


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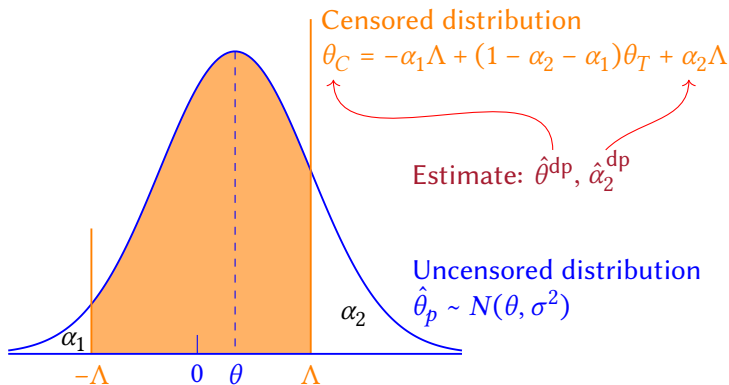


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Solve for θ (and σ^2, α_1)

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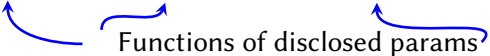
$$\hat{\theta}^{\text{dp}}(i), \hat{\alpha}_2^{\text{dp}}(i) \sim N \left(\begin{bmatrix} \hat{\theta}^{\text{dp}} \\ \hat{\alpha}_2^{\text{dp}} \end{bmatrix}, \begin{bmatrix} \hat{V}(\hat{\theta}^{\text{dp}}) & \widehat{\text{Cov}}(\hat{\alpha}_2^{\text{dp}}, \hat{\theta}^{\text{dp}}) \\ \widehat{\text{Cov}}(\hat{\alpha}_2^{\text{dp}}, \hat{\theta}^{\text{dp}}) & \hat{V}(\hat{\alpha}_2^{\text{dp}}) \end{bmatrix} \right)$$

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Functions of disclosed params



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- **Bias correction**: reduces bias *and* variance:

$$E(\tilde{\theta}^{\text{dp}}) \approx \theta, \quad V(\tilde{\theta}^{\text{dp}}) \lesssim V(\hat{\theta}^{\text{dp}})$$

Solving Political Problems Technologically

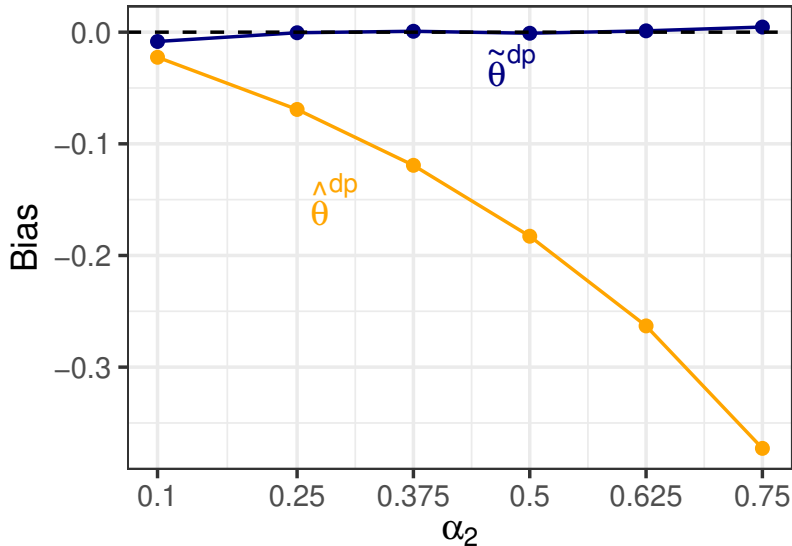
Differential Privacy & Inferential Validity

A General Purpose, Statistically Valid DP Algorithm

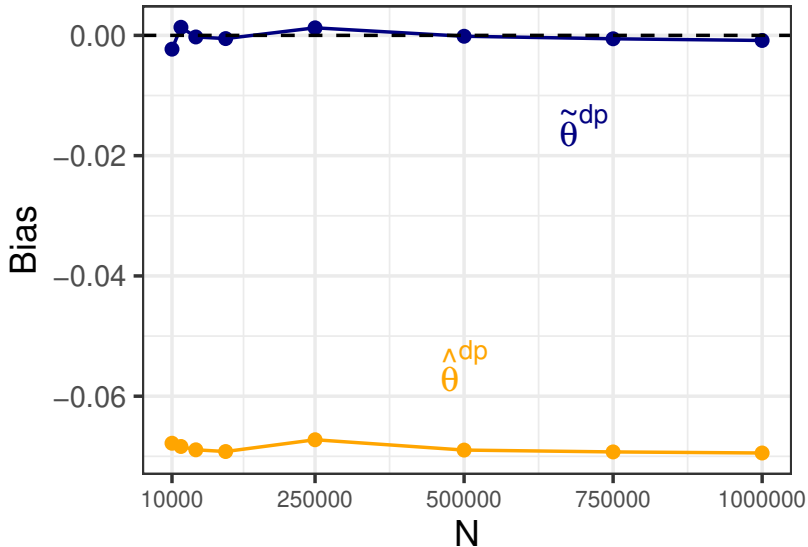
The Algorithm in Practice

Simulations: Finite Sample Evaluation

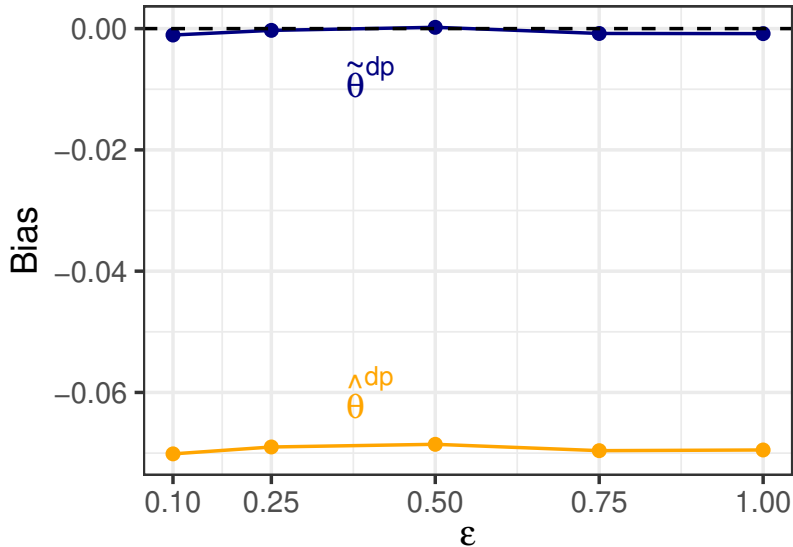
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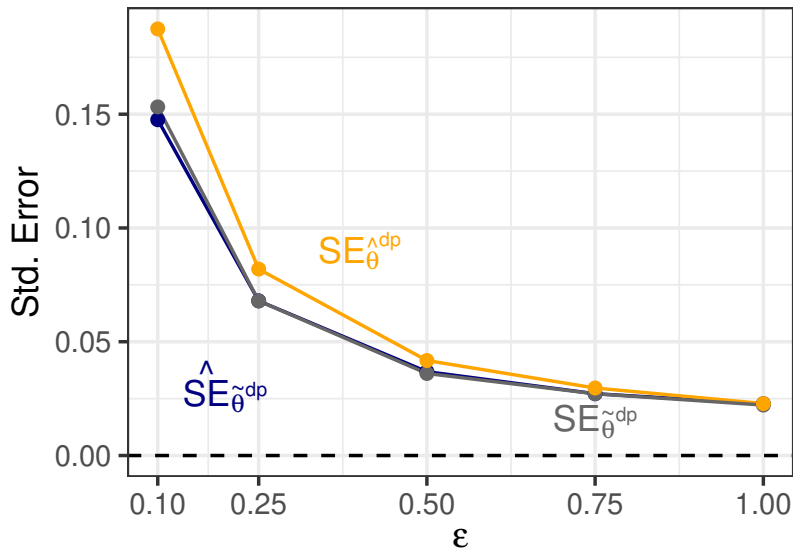
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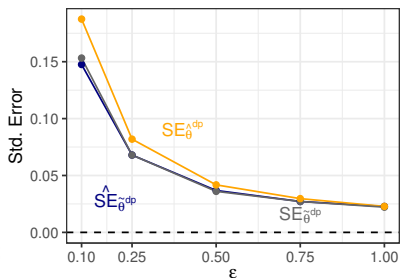
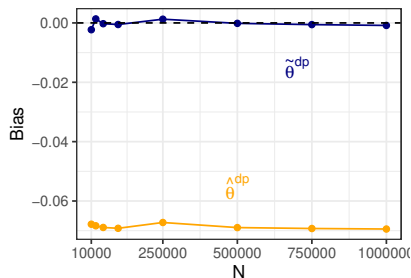
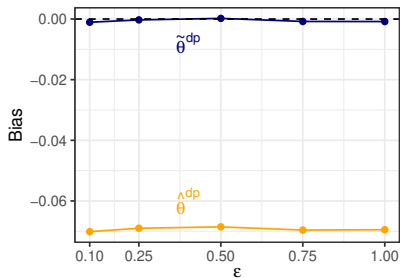
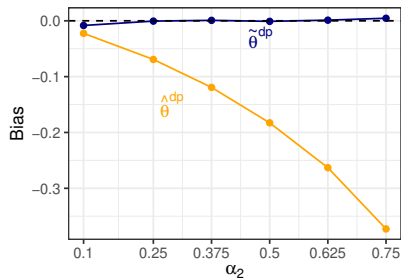
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For more information



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